P-P-121D
August 27, 1974
SUPERSEDING
Fed. Spec. P-P-121C
May 15, 1967

FEDERAL SPECIFICATION

PAPER, ABRASIVE, GARNET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers paper sheets and rolls, coated on one surface with garnet abrasive grain.

1.2 Classification.

1.2.1 Classes and Grit Sizes. Rolls and sheets shall be of the following classes and applicable grit sizes as specified (see 6.2).

Class 1 - Sheet, finishing, open coat.
Grit Sizes: 280, 240, 220, 180, 150, 120, 100, 80.

Class 2 - Sheet, closed coat.
Grit Sizes: 150, 120, 100, 80, 60, 50, 40, 36.

Class 3 - Roll, closed coat.
Grit Sizes: 220, 180, 150, 120, 100, 80, 60, 50, 40, 36, 24.

Class 4 - Sheet, open coat.
Grit Sizes: 150, 120, 100, 80, 60, 50, 40, 36.

Class 5 - Roll, closed coat, resin type.
Grit Sizes: 180, 150, 120, 100, 80, 60, 50, 40, 36.

Class 6 - Roll, open coat, glue type.
Grit Sizes: 120, 100, 80, 60, 50, 40.

Class 7 - Sheet, finishing, open coat, non-loading type.
Grit Sizes: 280, 240, 220, 180.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specification, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in

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the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification

MIL-P-3816 - Packaging and Packing of Abrasive and Abrasive Products for Shipment and Storage

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or requests for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 828 - Tensile Breaking Strength of Paper and Paperboard.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

Department of Commerce Product Standard:

PS 8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.)

Uniform Classification Committee, Agent:

National Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S Riverside Plaza, Chicago IL 60606.)

(Technical society and technical association specifications and standards are

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generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Garnet abrasive. The abrasive grain shall be crushed from clean, hard, abrasive grade garnet and shall not contain more than five percent of extraneous materials.

3.1.1.1 Grading control and tolerance. Grading of abrasive grain shall be in accordance with PS-8 and the tolerances therein shall apply to abrasive grain recovered from the coated abrasive product.

3.1.2 Backing and weight tolerance. The backing shall be a good quality rope or wood fiber paper suitable for the purpose intended and shall comply with the requirements of Tables I and II.

TABLE I. Weight of Backing

Classes	Grit Sizes	Nominal Backing Basic Weight 1/	Backing Weight Symbol or Designation
1 and 7	All	40	A weight, or finishing paper
2 and 4	150, 120, 100	70	C weight, or cabinet paper
2 and 4	80, 60, 50, 40, 36	90	D weight, or cabinet paper
3, 5, and 6	All	130	E weight paper

1/ The basic weight in pounds of 480 sheets 24 X 36 inches, with a tolerance of plus 5 percent, minus 7 percent, determined after conditioning for at least 24 hours at 70°F ± 2°F and 50 ± 2 percent relative humidity.

3.1.3 Adhesive. The adhesive for classes 1, 2, 3, 4, and 6 shall have a good quality animal hide glue base suitable for the purpose intended. The adhesive for class 5 shall have a good quality animal hide glue base for the making coat and hardenable synthetic resin for the sizing coat. The adhesive for class 7 shall have a good quality animal hide glue base plus a surface coating of metallic stearate or other material which will retard the filling of the abrasive while in use. As alternate practice classes 1, 2, 3, 4, and 6 may be furnished with a hardenable synthetic resin sizing coat.

3.2 Coating. Abrasive grain shall be applied to one side only of paper backing.

3.2.1 Open coat. Abrasive grain shall cover approximately 50 percent of the backing, thus leaving open spaces between the grains of the abrasive (see 6.1.3). Classes 1, 4, and 7 sheets and class 6 roll, shall be open coat.

3.2.2 Closed coat. Abrasive grain shall completely cover the surface of the backing (see 6.1.3). Class 2 sheet, and classes 3 and 5 roll, shall be closed coat.

3.3 Tensile breaking strength. The minimum average breaking strength shall be in accordance with Table II when tested in accordance with 4.4.2.

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TABLE II. Tensile Breaking Strength

Class	Designation	Tensile Strength, Average	
		Machine Direction	Cross Machine Direction
		Pounds/Inch, Min	Pounds/Inch, min
1 and 7	A	25	12
2 and 4	C	50	25
	D	65	32
3, 5, and 6	E	90	45

3.4 Form and dimensions.

3.4.1 Sheet. Unless otherwise specified (see 6.2) a sheet shall be 9 X 11 inch with a plus or minus tolerance of 1/8 inch in length, and 1/16 inch in width when measured after conditioning at 70°F plus or minus 2°F and 50 percent plus or minus 2 percent relative humidity for at least 24 hours.

3.4.2 Roll. A roll shall be 50 yards long, plus 12 and minus zero inches, and in nominal widths, as specified (see 6.2).

3.5 Identification marking. Every roll, 6 inches and wider, shall be marked at least once in every linear yard, and each sheet shall be marked in a legible and permanent manner with the following information:

(a) Manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.
(b) The name of the abrasive or the manufacturer's trade designation of such known character that the abrasive grain may be readily determined.
(c) The grit size representing the grain size.
(d) The backing weight designation.

3.6 Workmanship. Garnet abrasive paper shall be of the highest grade and equal or better in every respect to that which is considered good practice by industry and shall be free from imperfections which may affect its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. All abrasive paper of the same class and grain size manufactured under essentially the same conditions and offerred for delivery at one time shall be considered a lot for the purposes of examinations and tests.

4.2.2 Sampling for examination of packaging and packing. Packaged and packed abrasive paper in shipping containers shall be sampled in accordance with

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inspection level I.

4.2.3 Sampling for visual, dimensional, and workmanship characteristics of end item. Sampling for these characteristics shall be in accordance with inspection level S-4.

4.2.4 Sampling for tests. Sampling for test shall be selected in accordance with inspection level S-2.

4.3 Inspection and tests. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 Inspection.

4.3.1.1 Packaged and packed abrasive paper. Abrasive paper ready for shipment, selected in accordance with 4.2.2 shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 Individual sheets and rolls. Samples of abrasive paper selected in accordance with 4.2.3 shall be examined in accordance with Table III. The AQL's for defects listed in Table III shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.2 Lot acceptance tests. Samples selected in accordance with 4.2.4 shall be subjected to the tests in 4.4, AQL 6.5 percent defective.

TABLE III. Classification of Defects, End Item

Examine Applicable Paragraph	Defects	Major	Minor
1.2.1	Size of abrasive grit not as specified.	X	
3.1.1.1	Excessive extraneous material present in abrasive.	X	
3.2	Abrasive not closed or open coat as specified.	X	
3.4.1	Dimensions of sheets, not correct size or not within tolerances.	X	
3.4.2	Size of rolls not as specified.	X	
3.5	Identification marking omitted, illegible, incomplete, or incorrect.		X
3.6	Sheets and rolls torn or cracked.	X	

4.4 Tests.

4.4.1 Abrasive grain size.

4.4.1.1 Grading for abrasive grain size. Grading to determine the grain (grit) size, including recovery of grain from the sheets, preparation of grain for grading, and procedures shall be in accordance with the grading of grain as included in PS-8.

4.4.1.2 Certification of grading. Grading tests specified in 4.4.1.1 shall be omitted in instances where the contractor produces certification in writing that the abrasive grain complies with the requirements of PS-8. Contractors are urged to include this certification with material offerred for inspection, and

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inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 Breaking strength. The tensile breaking strength of the coated abrasive paper, directly after conditioning for at least 24 hours at a temperature of 21° ± 1°C (70° ± 2°F) and 50 ± 2 percent relative humidity, shall be determined by test method ASTM D 828, except that the jaws of the testing machine shall be 5 inches apart. Specimens 1 inch wide shall be taken both longitudinally and transversely from sample sheets and rolls. Undersize transverse specimens are permissible for narrow rolls. The average tensile strength in both directions shall be computed from at least three specimens. Samples tearing at the jaws should be discarded (see 3.3).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing of abrasive paper shall be in accordance with MIL-P-3816, as specified for level A, B, or C (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Abrasive paper designated "A" weight of backing is a lightweight flexible paper used chiefly for sanding by hand on finishing materials. Abrasive paper designated "C", "D", or "E" weight is a heavy, less flexible backing and is used chiefly for hand and machine sanding on wood or other surfaces where greater pressures are used for stock removal. Open coat garnet paper should be used for sanding soft wood which might clog the abrasive coating of closed coat grain.

6.1.1 Abrasive grit numbers. The number designations in the order given in 1.2.1 indicate the finest to the coarsest grit sizes.

6.1.2 Garnet. A group of hard minerals containing trisilicates of alumina, magnesia, calcia, ferrous oxide, manganese oxide, or chromic oxide. Almandite $3FeO.Al_2O_3.3SiO_2$ crystals of a deep-red color, is the most common garnet used for coating abrasive paper and cloth.

6.1.3 Coating. A closed coat is one in which the abrasive is put on in such quantity as entirely to cover the backing. An open coat covers about 50 percent of the backing thus leaving open spaces between the grains of abrasive.

6.1.4 Work value Coated abrasives should be stored in a cool even temperature where the relative humidity is between 40 to 50 percent. Best work value will be obtained ordinarily if stored and worked in rooms having a relative humidity of approximately 50 percent. Coated abrasives should not be stored on concrete floors.

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6.1.5 Belts. When coated abrasive belts are worked on wood having excessive moisture, or on glued wood which has not had sufficient time to dry thoroughly, it will tend to load the abrasive coating prematurely. Open coat is suggested for materials which tend to load.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class and grit size for sheet (see 1.2 and 3.4.1).
(c) Class, grit size, and width of rolls (see 1.2 and 3.4.2).
(d) Level of packaging and packing required (see 5.1).

Custodian:
Army - MR
Air Force - 84

User activities:

Army - MI, ME
Navy - YD, MC, OS, AS, SH

Preparing activity:
Air Force - 84

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P-P-121D, Paper, Abrasive, Garnet

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